Does OCT Enhancement Preserve What Clinicians Measure?

Lesion-level fidelity of retinal OCT enhancement — research proposal (draft)

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Background and motivation

OCT enhancement methods are judged by how the image looks, but clinicians act on what they measure. Denoising, super-resolution and generative restoration are routinely evaluated with PSNR and SSIM, yet treatment decisions in retinal disease rest on lesion measurements: fluid volume, the presence of small fluid pockets, hyperreflective foci.

Our previous work (OMIA 2026) showed this gap is real for fundus imaging: generative restoration improved pixel metrics while degrading downstream diagnostic accuracy (Spearman $\rho = -0.50$). Whether the same holds for OCT is unknown, and OCT is a harder case. Its main degradation is speckle, a physical interference pattern rather than additive noise, and the lesions that matter can be only a few pixels across.

If enhancement erases a small fluid pocket or invents one, the error passes silently into the measurement a clinician relies on. This project asks whether that happens, how often, and whether it can be detected without ground truth.

Research question

Main question: Does retinal OCT enhancement preserve the lesion information clinicians measure, and can we detect, without ground truth, when it does not?

Sub-questions:

1. **Fidelity.** How do denoising, super-resolution and generative enhancement change lesion segmentation and lesion volume compared with the original scan?
2. **Small lesions.** Are small lesions disproportionately erased or hallucinated, and does this depend on lesion size and type?
3. **Metric validity.** Do standard image-quality metrics (PSNR, SSIM, CNR) predict lesion-level fidelity?
4. **Detection.** Can a no-reference score flag scans whose diagnostic content was damaged by enhancement?

5. **Backbone (exploratory).** Is a self-supervised, foundation-model segmenter more or less robust to enhancement than one trained from scratch?

Hypotheses

ID	Hypothesis	How it is tested	Supported if
H1	Enhancement improves image-quality metrics but does not improve lesion fidelity; generative methods degrade it.	Lesion Dice and volume error on enhanced vs original scans, per method	Quality metrics rise while volume error does not fall, and rises for generative methods
H2	Fidelity loss is concentrated in small lesions.	Detection rate and volume error stratified by lesion size bin	Error increases as lesion size decreases
H3	Image-quality metrics are weak or negative predictors of lesion fidelity.	Rank correlation between PSNR/SSIM/CNR and lesion volume error across scans and methods	Correlation near zero or negative
H4	A no-reference consistency score flags scans damaged by enhancement.	AUROC for detecting scans with large volume error, using only the original and enhanced images	AUROC clearly above chance and above quality-metric baselines
H5 (exploratory)	An SSL-pretrained segmenter is more robust to enhancement than one trained from scratch.	Compare fidelity loss between an SSL-pretrained segmenter and nnU-Net	Smaller fidelity loss for the SSL model

A negative result on H1 (enhancement preserves lesions) is still publishable if H2 or H3 hold, since it would bound when enhancement is safe.

Aims and objectives

The aim is to measure the lesion-level cost of OCT enhancement and give practitioners a way to detect it.

1. Build a reproducible benchmark that applies several enhancement methods to public retinal OCT with lesion labels.
2. Quantify lesion fidelity with clinical endpoints: volume error, small-lesion detection, erased and hallucinated lesions.
3. Test whether standard image-quality metrics predict that fidelity.
4. Propose and validate a no-reference check that flags scans whose lesion content changed.
5. Release the code and benchmark so others can evaluate new enhancement methods against lesion fidelity.

Methods

The design crosses enhancement methods with segmenters and scores every combination against lesion labels on the original scans.

Data. Public retinal OCT with pixel-level lesion labels. Primary candidate: RETOUCH (intraretinal fluid, subretinal fluid, pigment epithelial detachment; three vendors). Secondary candidates to confirm during the data survey: Duke DME and other sets with fluid or hyperreflective-foci labels. Unlabelled OCT for SSL pretraining comes from larger public collections.

Enhancement methods (3–4):

Family	Example	Why include it
Classical	BM3D or similar despeckling	Non-learned reference
Supervised CNN	A denoiser trained on averaged B-scans	Common in practice
GAN	A published OCT GAN denoiser	Known to hallucinate texture
Diffusion	A diffusion-based restoration model	Strongest pixel metrics, highest hallucination risk

Published pretrained weights are preferred, so the study tests methods as others would use them.

Segmenters (2): nnU-Net trained from scratch, and a segmenter built on an SSL foundation backbone (e.g. RETFound). Both are trained on original scans and then run on enhanced versions, so any change in output is caused by the enhancement.

Endpoints:

- Image quality: PSNR, SSIM, and a no-reference measure such as CNR
- Segmentation: Dice per lesion type
- Clinical: absolute and relative lesion volume error
- Small lesions: detection rate by size bin; counts of erased and hallucinated lesions

Fidelity check (H4). A no-reference score computed from the original and enhanced scans only, for example the disagreement between segmentations of the two, or the segmenter's uncertainty change. It is evaluated by AUROC for flagging scans above a clinically meaningful volume-error threshold.

Analysis. Rank correlation for H3, paired tests across methods for H1 and H2 with correction for multiple comparisons, and bootstrap confidence intervals throughout.

Expected contributions

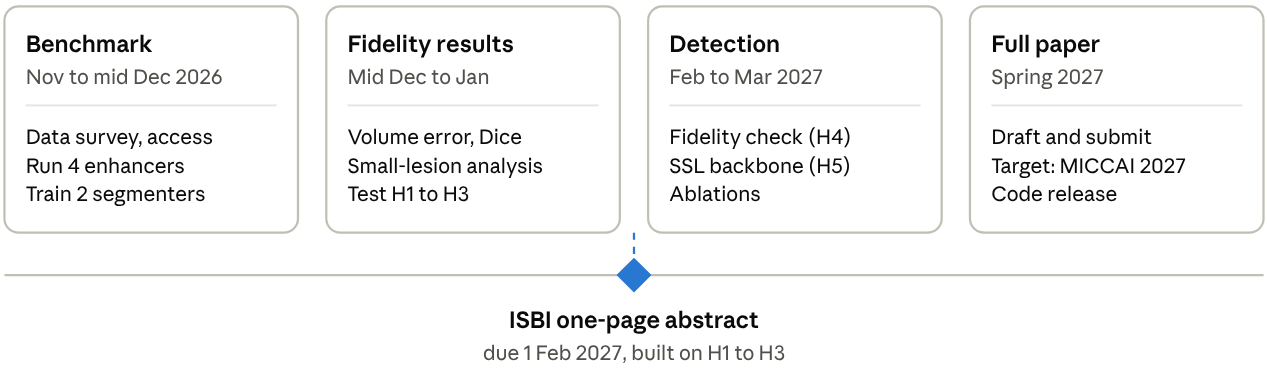
The contribution is a finding plus a fix, not a new segmentation architecture.

1. The first systematic measure of lesion-level fidelity across OCT enhancement families, using clinical endpoints rather than pixel metrics.
2. Evidence on whether small, high-stakes lesions are the ones enhancement erases or invents.
3. A no-reference check that flags damaged scans, usable before an enhanced image reaches a clinician or a downstream model.
4. An open benchmark for judging future enhancement methods by what they preserve.

How this differs from earlier work. The OMIA 2026 paper studied fundus images and diagnostic grading. This project changes the modality (3D OCT with speckle), the endpoint (lesion volume and small-lesion detection) and adds a detection method. It is kept separate from the MIDL 2027 thesis extension, which covers lesion work on fundus data.

Timeline and target venues

First results land in time for the ISBI abstract on 1 February



project roadmap · 4 phases, 1 gate

The ISBI 2027 abstract (Lausanne, 25–28 May) carries the first fidelity results; it is unpublished, so the full paper targets MICCAI 2027, with the OMIA 2027 workshop as the fallback. Confirm the MICCAI deadline once it is announced.

Risks and resources

Risk	Mitigation
Reviewers see it as the OMIA paper on a new modality	Lead with the OCT-specific physics, the volume endpoint and the detection method
Lesion labels exist only on original scans	Treat the original-scan labels as the reference; this is the clinically relevant comparison
Pretrained enhancement models are hard to obtain or run	Start with BM3D and one learned denoiser; add GAN and diffusion as access allows
Effects are small or inconsistent	A null result bounded by H2 and H3 is still reportable; stratify by lesion size to find where effects concentrate
Compute limits on a laptop	Prototype on 2D B-scans locally; run the full grid on Colab or university GPUs

Resources: MacBook Air M5 for prototyping, cloud or university GPU time for the full experiment grid, public datasets only (no ethics approval needed).